BrandGuard AI — Round 3 Final Build & Hardening

TechCircuit Hackathon • Team TECHSPARKZ • Tagline: *Create Freely. Publish Confidently.*

1. Executive Summary

In Round 3 of the TechCircuit Hackathon, **BrandGuard AI** was advanced into a production-grade, hardened, multi-tenant brand governance platform. The platform serves as an enterprise pre-publication compliance firewall, combining automated deterministic rule execution, OCR-based visual canvas verification, strict organization-level data isolation, OWASP-standard Argon2id password security with transparent on-login migration, and human-in-the-loop review cockpit.

Capability	Round 2 Baseline	Round 3 Production Hardening
Password Security	Salted SHA-256	OWASP Argon2id with automatic constant-time migration
Multi-Tenancy	Role-based permissions	Strict organization isolation (org-novatech, org-apex) with 403 bounds
Visual Review	Palette & contrast only	Tesseract OCR canvas text extraction + WCAG 2.2 AA contrast validation
Workflow & Gatekeeping	Approve / Change Request	Rejection endpoint + Critical violation approval blocking + Separation of duties
Governance Insights	Global counts (duplicates)	Multi-tenant dynamic filtering + Largest-remainder 100% sum rule
Automated Tests	12 tests passing	32 automated tests passing (100% pass rate in 5.40s)

2. Implemented Engineering Improvements

A. OWASP Argon2id Password Security & Transparent Migration

- **Modern Cryptographic Standard:** Integrated argon2-cffi with parameters: time_cost=3, memory_cost=65536 (64 MiB), parallelism=4, hash_len=32, salt_len=16.
- **Zero-Disruption Migration:** Automatically detects legacy SHA-256 hashes on authentication, verifies in constant time, and seamlessly upgrades the database record to Argon2id without requiring password resets.
- **Security Guarantees:** Plaintext passwords are never cached, logged, or serialized.

B. Organization-Level Multi-Tenant Data Isolation

- **Domain Separation:** Configured isolated organizations (org-novatech and org-apex). Brands, approved facts, content items, and audit logs are strictly partitioned.
- **Server-Side Gatekeeping:** Enforced validate_org_access across all endpoints. Cross-organization access attempts immediately return HTTP 403 Forbidden.
- **Separation of Duties:** Creators are strictly prohibited from approving content or approving their own submissions. Reviewers and admins execute human-in-the-loop decisions.

C. OCR Visual Brand Analysis & Accessibility

- **Canvas Text Extraction:** Tesseract OCR engine extracts text embedded within promotional image assets (PNG, JPEG, WebP).

• **Graphic Asset Governance:** Scans extracted image text against prohibited terminology, pricing mismatches, and unsubstantiated absolute claims.

• **WCAG 2.2 AA Contrast:** Computes relative luminance and color contrast ratios against the 4.5:1 standard with graceful offline fallbacks.

D. Governance Insights & Workflow Deduplication

• **Root Cause Deduplication:** Eliminated duplicate drafts through backend draft deduplication, campaign reference lookup, and revision cleanup.

• **Dynamic Scoping:** All metrics and frequent mistake patterns dynamically filter by active organization and brand.

• **Largest-Remainder Rule:** Issue category breakdown percentages always sum to exactly 100.0%.

• **7-Column Recent Activity Audit Table:** Content Title, Brand, Submission Date, Channel, Status, Revision, Last Updated.

3. Automated Test Verification (32/32 Passed — 100%)

Test Suite	Tests	Execution Time	Status
Core Engine (test_brandguard.py)	12 Scenarios	1.42s	**PASSED**
Authentication & RBAC (test_round3.py)	8 Scenarios	1.18s	**PASSED**
Hardening & Workflow (test_round3_hardening.py)	12 Scenarios	2.80s	**PASSED**
Total Test Suite	**32 Scenarios**	**5.40s**	**100% PASS RATE**